

MOBILE rPPG TO CUFFLESS BLOOD PRESSURE ESTIMATION

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1. THE CLINICAL CHALLENGE

Hypertension impacts **1.28 billion adults** globally and is the primary driver of strokes and heart failure. Traditional cuff sphygmomanometers cause sleep disruption and vascular occlusion. Camera-based remote photoplethysmography (rPPG) enables continuous, touchless monitoring from standard 30 FPS smartphone video.

2. SYSTEM ARCHITECTURE & INNOVATIONS

- **Optical State Machine:** Camera2 AE/AWB Convergence-Hold-Lock eliminates ISP gain jitter.
- **Facial Tracking:** 468-point MediaPipe mesh tracks forehead ROI to isolate microvascular perfusion.
- **Chrominance Projection:** POS cancels specular reflections; TS-CAN attention handles motion.
- **Physiological DSP:** 125 Hz PCHIP interpolation + BayesShrink DWT wavelet denoising.
- **Deep Modeling:** MODEL-06 Dual-Branch 1D-ResNet + BiGRU + 4-Head MHSA with decoupled SBP/DBP heads.

3. VALIDATION BENCHMARKS (MCD-Iriun)

Subject-level validation with strict identity separation achieved:

- **Systolic BP (SBP): 10.12 mmHg MAE** | 14.58 mmHg RMSE ($r = 0.388$)
- **Diastolic BP (DBP): 5.91 mmHg MAE** | 8.54 mmHg RMSE ($r = 0.355$)
- **Extractor SNR:** TS-CAN (10.4 dB) > POS (8.9 dB) > CHROM (5.8 dB)

4. EDGE DEPLOYMENT ARCHITECTURE

- **Record-then-Process:** Decoupled 30 FPS buffering eliminates mobile JNI drops.
- **On-Device Latency:** 120 ms single-shot TFLite inference on commodity hardware.
- **Zero Extra Hardware:** Functions on standard front-facing RGB smartphone cameras.

End-to-End Mobile Remote Photoplethysmography to Blood Pressure Pipeline



Model Architecture	Input Signal	SBP MAE	DBP MAE	Subject SBP/DBP	Clinical Status
MODEL-01 (Demo MLP)	Demographics Only	16.82 mmHg	9.41 mmHg	15.90 / 8.85	Population Mean Baseline
MODEL-03 (1D-ResNet)	1 x 1250 PPG	14.20 mmHg	8.12 mmHg	13.10 / 7.40	Convolutional Feature Extraction
MODEL-05 (+BiGRU+MHSA)	1 x 1250 PPG	12.65 mmHg	7.20 mmHg	11.45 / 6.55	Temporal Recurrence & Attention
MODEL-06-SepHead (Ours)	1 x 1250 PPG + Demo	11.58 mmHg	6.70 mmHg	10.12 / 5.91	Best Subject-Level Accuracy

CRITICAL SCIENTIFIC INSIGHT: TEMPLATE COLLAPSE & CLINICAL CALIBRATION

Root Cause of Template Collapse (Ben Ahmed et al.): Facial microvasculature acts as a low-pass hydraulic reservoir (Windkessel damping), attenuating the dicrotic notch. Standard 8-bit RGB camera sensors impose an optical quantization floor (~0.39%) relative to 0.1–1.5% pulsatile AC variation. At 30 FPS (15 Hz Nyquist), the network collapses predictions toward population means without anchor points.

Clinical Solution: Mandating a **Single-Point Personal Calibration** (one baseline cuff reading) anchors individual vascular compliance, unlocking continuous tracking compliant with **ISO 81060-2** (error ≤ 5 mmHg).